Language and Earnings: The Association Between Chinese Language Use and Wages in the United States

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1 Introduction

In 2006 and 2010, Genifer and Kerry Stewart enrolled their two Black sons into Wesley International Academy, one of the only public elementary schools in the greater-Atlanta area that offered/required instruction in Mandarin Chinese. I was the second of the sons. And, from that point on, I have developed a deep appreciation for Chinese culture and history, as well as a mediocre grasp of the language.

A few years ago I finally asked my parents to explain their motive, which was simple: With China's growing economic, political, and social influence, they believed their children should be equipped with the skills to solve the problems the future demands — one of which being speaking a Chinese language.

16 years later, the present study examines this presumption using data: does Chinese language ability translate into a measurable economic advantage in the United States?

In particular, this paper addresses the following research question: *Is Chinese language use associated with wage differences in the United States?* The present study is a descriptive study, aiming to document associations in observational data, rather than establish causation or predict out-of-sample outcomes. Although the other two approaches could yield fruitful data in this context, I argue that descriptive analysis is most appropriate here because, despite Chinese being one of the fastest growing languages in the world and China economically rivaling the United States, the relationship between Chinese language use and wages specifically in the American context is not thoroughly examined in existing literature.

The present study uses data from the Integrated Public Use Microdata Series (IPUMS) American Community Service (ACS), particularly analyzing samples from 2000-2004 and 2020-2024. Pre-analysis, I expected the following results: (1) Chinese speakers would earn more than non-Chinese speakers, especially in more recent years, (2) after controlling for demographic variables, the positive association would shrink but still exist, and (3) non-ethnically Chinese Americans with Chinese language fluency would earn more than ethnically Chinese Americans with Chinese language fluency.

The primary findings support the first hypothesis and partially supports the third hypothesis, while rejecting the second. Instead of seeing a small but positive association between language fluency and wage after controlling for demographic variables, the assocaition actually flips negative after controlling for race. I argue through subsequent analysis that this reversal in association suggests that initial association reflects those who speak Chinese, which are predominantly recent immigrants or Asian Americans with labor market, who may face labor market disadvantages, instead of the actual returns of language fluency itself.

2 Data

2.1 Replication

The raw IPUMS ACS extract (data1.csv.gz) is approximately 5.1 GB, which is too large to include in the GitHub repository.

To replicate this set, go to https://usa.ipums.org and create an new extract with the following:

Samples : 2000 5%, 2001–2004 ACS, 2020–2024 ACS (10 samples total)

Variables: YEAR, SAMPLE, SERIAL, CBSERIAL, HHWT, CLUSTER, STRATA, GQ, PERNUM, PERWT, FAMSIZE, SEX, AGE, RACE, RACED, HISPAN, HISPAND, BPL, BPLD, ANCESTR1, ANCESTR1D, ANCESTR2, ANCESTR2D, YRNATUR, YRIMMIG, YRSUSA1, LANGUAGE, LANGUAGED, SPEAKENG, RACASIAN, RACBLK, SCHLTYPE, EMPSTAT, EMPSTATD, INCTOT, INCWAGE

Format : .csv, Rectangular (person)

After placing the downloaded file relative to your .Rmd, run the chunk below to generate the .rds used in this analysis:

```
data_raw <- fread("raw/data1.csv.gz",
                  select = c("YEAR","INCTOT","LANGUAGE","AGE", "SEX","RACE","EMPSTAT"))
saveRDS(data_raw, "loaded_data_partial.rds")
```

Both the raw and filtered files (5.1 GB and 134.4 MB, respectively) both exceed what can be uploaded to GitHUb.

2.2 Filtering the Sample

The present study utilizes publicly available data from the IPUMS ACS, which is a nationally representative survey conducted by the U.S. Census Bureau that provides extremely detailed information on the demographic and economic characteristics of millions of American citizens. For this analysis, I pooled annual samples from 2000-2004 and 2020-2024, resulting in a combined dataset of over 34 million observations before filtering.

The raw data set combined ten samples from 2000-2004 and 2020-2024, but given technological constraints, this analysis focused on only the following variables for the 34 million total respondents: YEAR, INCTOT (income), LANGAUGE (language(s) spoken at home), AGE, SEX, RACE, and EMPSTAT (employment status).

In particular, to filter the limited sample to only the economically active population, I restrict the sample to employed indivdiuals (EMPSTAT = 1 or 2) between the ages of 22 and 64 with positive total income (INCTOT > 0). Filtering reduces the dataset to approximately 14 million observations. Again due to technological constraints, I then draw a random sample of 500,000 observations (seed = 1) and conduct all analyses on this subsample. Within the sample, 5,412 individuals (approximately 1.1%) speak Chinese at home.

```
data <- readRDS("loaded_data_partial.rds")


data <- data[EMPSTAT %in% c(1, 2) & INCTOT > 0 & AGE >= 22 & AGE <= 64]


data[, race_label := factor(RACE,
                            levels = 1:9,
                            labels = c("White",
                                       "Black",
                                       "American Indian/Alaska Native",
```

```
                                        "Chinese",
                                        "Japanese",
                                        "Other Asian/Pacific Islander",
                                        "Other race",
                                        "Two races",
                                        "Three+ races"))]

data[, sex_label := factor(SEX,
                           levels = c(1, 2),
                           labels = c("Male",
                                      "Female"))]

data[, chinese := ifelse(LANGUAGE == 43, 1, 0)]
data[, log_income := log(INCTOT)]
data[, period := ifelse(YEAR <= 2004,
                        "2000-2004",
                        "2020-2024")]

set.seed(1)
data <- data[sample(.N, 500000)]
```

2.3 Key Variables

2.4 Descriptive Statistics

3 Analysis

4 Conclusion

4.1 Limitations

This study was also limited purely by technology. Although additional variables, such as YRIMMIG (the year the respondent immigrated to the United States), BPL (birthplace), SCHLTYPE (whether the respondent attended public or private school), and ANCESTR1 (the respondent's self-reported ancestry or ethnic origin), could be beneficial to further analysis, technological constraints required that they be excluded from the sample.

5 References

Steven Ruggles, Sarah Flood, Matthew Sobek, Daniel Backman, Grace Cooper, Julia A. Rivera Drew, Stephanie Richards, Renae Rodgers, Jonathan Schroeder, and Kari C.W. Williams. *IPUMS USA: Version 16.0* [dataset]. Minneapolis, MN: IPUMS, 2025. https://doi.org/10.18128/D010.V16.0